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Master's Thesis

Adaptive Stress Management: A Human-in-the-Loop Framework using Active Learning and Physiological Computing

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Abstract

Stress significantly impacts physical and mental health, but inherent inter-subject variability makes static stress detection models unreliable for general use. This thesis proposes an Adaptive Stress Management system that uses a Human-in-the-Loop (HITL) framework combining real-time physiological computing with stream-based Active Learning (AL). The system is built on a distributed Robot Operating System 2 (ROS 2) architecture, processing Electrodermal Activity (EDA) and Blood Volume Pulse (BVP) signals. To address subject-wise variability, an active learning mechanism monitors prediction entropy and triggers a Large Language Model (LLM) based dialogue manager to query the user when the model's uncertainty is high.

Evaluations show the system operates well within acceptable real-time latency margins, processing live hardware streams at an average rate of 295.50 ms. Incorporating user feedback via active learning significantly increases stress detection accuracy from 64.2% to 88.0% and effectively reduces redundant system interruptions. Furthermore, the Qwen2.5:3B LLM intent classifier outperforms traditional keyword-based rule models in mapping ambiguous user responses, achieving 80.69% accuracy versus 48.07% on a custom dataset. Finally, while SHapley Additive exPlanations (SHAP) are not yet integrated into the system and are planned for future work, a pilot user study where users were shown two scenarios, one with and the other without explained queries, resulted in a 72.7% user preference for explained queries, demonstrating increased user acceptance.

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1

Introduction

1.1 Motivation

Stress significantly affects an individual's mental and physical health, reducing overall well-being and productivity. While effective management is crucial, individuals often struggle to recognize or address stress in a timely manner. Advances in technology offer solutions by leveraging physiological signals, measurements of biological functions such as heart rate (HR) and electrodermal activity (EDA); to detect and mitigate stress in real time [1]. These signals provide objective, involuntary measurements of stress responses, making them more reliable and difficult to manipulate than self-reported assessments [2, 3].



Figure 1.1: Common physiological sensors and wearable devices [4]

However, a primary challenge in this field is inter-subject variability. Physiological responses to stress differ significantly between individuals; what constitutes a high-stress signal for one person may represent a baseline state for another. Traditional static models often fail to generalize to new users without extensive labeled data. To address this, a Human-in-the-Loop (HITL) system for personalized stress monitoring utilizing an Active Learning framework is developed in this project. The system monitors

physiological data and upon detecting predictive uncertainty, proactively queries the user for clarification. This interaction is facilitated by a Large Language Model (LLM) to offer natural and empathetic inquiries, ensuring interventions are personalized and uncertainty-aware.

1.1.1 Project Relevance

The results of this project primarily benefit individuals seeking accessible, real-time mental health support, particularly those who are socially isolated or living alone. Loneliness exacerbates stress and is increasingly prevalent [5]. In the United States, nearly 30% of households are single-person households, and in the United Kingdom, single-person households have grown by 8% over the past decade [6, 7]. This demographic often lacks immediate social support networks to identify stress. An empathetic dialogue agent fills this gap by providing check-ins that feel like a supportive peer rather than a clinical tool [8, 9].

Existing solutions like Headspace or Calm [10, 11] rely on pre-scheduled activities and self-initiation. They lack real-time detection, often missing the critical window to address stress as it occurs, and employ generic approaches that fail to adapt to user baselines. By implementing active learning, this system reduces error rates, translating to fewer false-positive interruptions and higher precision interventions [12]. This transforms stress management from a reactive manual task into a proactive, automated safety net.

1.2 Challenges and Difficulties

A core difficulty in physiological stress detection is the inherent variance in autonomic nervous system responses across different individuals. Factors such as baseline electrodermal activity, resting heart rate, age, and fitness levels mean that fixed global modals yield high error rates when applied universally. Developing a scalable system requires models that can rapidly adapt to individual user baselines without requiring exhaustive offline calibration datasets. Wearable physiological monitoring systems frequently suffer from false-positive alerts driven by ambient noise, physical movement, or mild physiological arousal. When a monitoring system repeatedly interrupts a user with inaccurate stress alerts, it induces alert fatigue. Minimizing redundant interruptions during prolonged physiological episodes is critical to maintaining user acceptance and long-term engagement. Integrating continuous quantitative biosignals (such as EDA and HR) with qualitative human interaction presents a significant engineering challenge. Translating numerical uncertainty into context-aware, empathetic natural language inquiries, while accurately parsing complex user responses that contain temporal shifts, physical symptoms, or ambiguity; requires robust semantic classification beyond simple keyword rule matching.

1.3 Problem Statement

Current digital mental health interventions are predominantly reactive, relying on manual user initiation and static detection rules that ignore individual physiological differences. Wearable sensors capture continuous autonomic data, but existing deployment pipelines struggle with noise handling, high false-positive rates, and rigid communication formats. This research addresses these limitations by establishing a real-time, closed-loop framework combining ROS 2 physiological streaming, uncertainty-

driven active learning, and intent-aware LLM dialogue management to deliver personalized, adaptive stress support.

1.3.1 Research Questions

To address the gaps identified in the literature and guide this study, the following research questions have been formulated:

RQ1: Can a real-time streaming architecture maintain comparable stress detection performance when transitioning from streams of pre-processed data to raw, noisy physiological signals?

- **Null Hypothesis ($H_{0,1}$):** There is a significant difference in stress detection performance when the input source switches from pre-processed dataset streams to raw, real-time sensor feeds.
- **Alternative Hypothesis ($H_{1,1}$):** There is no significant difference in model performance when the input source switches from pre-processed dataset streams to raw, real-time sensor feeds.

RQ2: Can an active learning framework reduce the frequency of redundant interruption during prolonged physiological episodes compared to a static model?

- **Null Hypothesis ($H_{0,2}$):** There is no statistically significant difference in the redundant interruption rate between a static model with fixed uncertainty rules and an adaptive active learning model.
- **Alternative Hypothesis ($H_{1,2}$):** The adaptive active learning model significantly reduces the redundant interruption rate compared to the static model.

RQ3: To what extent active learning with uncertainty sampling improve the personalization accuracy of stress detection compared to a static, subject-independent model?

- **Null Hypothesis ($H_{0,3}$):** There is no significant difference in stress detection accuracy between a static, subject-independent model and an active learning model with uncertainty sampling.
- **Alternative Hypothesis ($H_{1,3}$):** An active learning model with uncertainty sampling achieves significantly higher personalization accuracy and lower false-positive rates compared to a static, subject-independent baseline by progressively updating on user-specific data.

RQ4: Does the use of intent-aware LLM queries improve the correctness of mapping ambiguous user responses to ground-truth stress labels compared to keyword-based rule systems?

- **Null Hypothesis ($H_{0,4}$):** There is no statistically significant difference in mapping accuracy between the LLM-based intent classifier and the keyword-based rule system.
- **Alternative Hypothesis ($H_{1,4}$):** The LLM-based intent classifier achieves significantly higher mapping accuracy than the keyword-based rule system.

RQ5: Can an uncertainty-based query with a natural language explanation of the physiological prediction increase user acceptance compared to unexplained queries?

- **Null Hypothesis ($H_{0,5}$):** Providing a natural language explanation for a physiological prediction (e.g., explaining why a query was triggered) does not significantly alter user acceptance in the system.
- **Alternative Hypothesis ($H_{1,5}$):** Users exposed to inquiries accompanied by natural language explanations of physiological triggers report significantly higher acceptance compared to users exposed to unexplained queries.

2

Background

To ensure clarity throughout this report, this section defines the core concepts and technologies foundational to the proposed framework.

2.1 Stress Detection using physiological signals

Physiological signals are the involuntary physical changes our bodies make, such as heart rate, breathing, skin temperature, and sweating. These signals are very helpful for detecting stress because they provide a direct window into the human nervous system. The body regulates stress through two main systems. The Sympathetic Nervous System (SNS) is responsible for the body's "fight or flight" response [13]. When a person is stressed, the SNS activates, causing the heart to beat faster and the skin to sweat. On the other hand, the Parasympathetic Nervous System (PNS) controls the "rest and digest" functions. When a person is relaxed, the PNS slows down the heart rate and calms the body. By measuring physiological signals, wearable sensors can see which system is active and accurately detect if a person is stressed or relaxed. To measure the body's stress response, this study focuses on two specific signals:

- **EDA (Electrodermal Activity):** This signal measures tiny changes in how much a person is sweating. Even microscopic amounts of sweat, which a person might not even feel, change the electrical conductivity of the skin. Because sweating is controlled directly by the SNS, EDA is one of the most reliable indicators of stress [14].
- **IBI (Inter-Beat Interval):** While heart rate measures how many times the heart beats in a minute, IBI measures the exact time (in milliseconds) between each individual heartbeat. Measuring these tiny variations helps detect stress.

IBI is often derived using a PPG (Photoplethysmogram) or BVP (Blood Volume Pulse) sensor [14]. These sensors work by shining a light into the skin to measure blood flow. Every time the heart pumps, a wave of blood rushes through the wrist, changing how much light is reflected back to the sensor. By tracking the peaks of these blood flow waves, the system can calculate the exact time between each heartbeat to find the IBI.

2.2 Human-in-the-Loop (HITL)

HITL is a design approach where an Artificial Intelligence (AI) system does not make all the final decisions completely on its own. Instead, it keeps a human involved in the process. This is useful because AI models are not always 100% accurate, especially when trying to guess complex human emotions like stress. By keeping a human-in-the-loop, the AI can ask the user for feedback when it is unsure. This prevents the system from making frustrating mistakes, making the technology safer, more reliable, and more trustworthy.

2.2.1 Active Learning

Active Learning is a specific machine learning technique where the AI actively asks the user for the correct answer when it is confused. It is needed because every person experiences stress differently, so a generic one-size-fits-all model usually performs poorly. Instead of needing thousands of pre-labeled examples for every single user, an active learning system learns on the fly by asking the user directly [15].

Example: If a wearable sensor detects a sudden spike in heart rate, the AI might not know if the user is feeling stressed or if they are simply walking up a flight of stairs. Instead of randomly guessing "stress" and triggering a false alarm, an active learning system will send a quick prompt asking: "Your heart rate went up, are you feeling stressed?" Once the user answers, the AI updates its own rules so its predictions get better for the next time.

Based on how the data is accessed, active learning environments are generally categorized into three distinct settings [15, 16]:

- **Membership Query Synthesis:** The learner requests labels for any unlabeled instance in the input space, including entirely new, artificially generated queries.
- **Stream-Based Selective Sampling:** Also known as sequential sampling, unlabeled instances are drawn one at a time from a continuous data source. The system must decide instantaneously whether to query the human or discard the sample.
- **Pool-Based Sampling:** The entire collection of data is evaluated at once and ranked to select the best overall elements for human annotation.

To determine exactly when to prompt the human for a label, systems rely on specific querying strategies [15, 16]:

- **Uncertainty Sampling:** The model selects samples for which it has the least statistical confidence, prompting humans to provide labels for those specific data points.
- **Query-by-Committee:** Multiple models or hypotheses are used, and the samples on which these models disagree the most are selected for human annotation.
- **Diversity Sampling:** This strategy targets unannotated data points that are underrepresented or entirely absent from the existing training set, thereby broadening the model's overall coverage of the feature space.

In this thesis, a stream-based selective sampling approach paired with entropy-based uncertainty sampling is utilized. Because physiological data from wearable sensors arrives continuously in real-time, the framework cannot rely on a static pool of data. Instead, the system must evaluate each physiological data window sequentially, calculating its entropy to decide instantaneously whether to trigger a Human-in-the-Loop query or proceed with an automated intervention.

2.2.2 Static vs. Adaptive Models

When building AI systems, models generally fall into two categories based on how they learn:

- **Static Model:** A static model is trained once using a dataset and then locked in place. Its rules never change after it is deployed. While this is easier to build, it relies on a one-size-fits-all approach. For example, a static model might assume that a heart rate of 100 beats per minute always means a person is stressed. It will apply this exact same rule to every person forever, which might cause false alarms for someone who naturally has a higher resting heart rate.
- **Adaptive Model:** An adaptive model continuously updates and changes its rules over time based on new data or direct user feedback. Instead of staying exactly the same, it personalizes itself to the specific user. For example, if the user repeatedly tells the system, "No, I am not stressed right now" when their heart rate reaches 100, the adaptive model will learn this personal baseline and adjust its rules to stop sending false alarms in the future.

2.3 Natural Language Processing and Intent Classification

When humans interact with an AI, the system needs to understand what the person actually means. In Natural Language Processing (NLP), this is called Intent Classification. It is the process of reading a user's text or speech and sorting it into a specific category or intent (such as asking for help, confirming a feeling, or telling the system to pause).

Traditionally, systems used Keyword-Based Models. These are simple rules that look for exact words. For example, if a user types "stress", the system flags them as stressed. However, keyword systems break easily. If a user types, "I am completely overwhelmed," a keyword system will fail to detect stress because the word "stress" is missing.

To solve this, modern systems use Large Language Models (LLMs). LLMs are advanced AI models that understand the underlying meaning and context of a sentence, rather than just looking for exact words. An LLM understands that "I am overwhelmed" or "I am stressed" means a similar thing, making the system natural to talk to.

2.3.1 Context-Awareness in Real-Time Systems

For a real-time system, understanding exactly when and who the user is talking about is critical. A context-aware system prevents the AI from making silly mistakes. Two common problems an LLM helps prevent are:

- **Past Misclassifications:** If a user says, "I was very stressed yesterday", a keyword model might just see the word "stressed" and trigger an alarm. A context-aware LLM understands this happened in the past and knows not to trigger a real-time intervention today.
- **External Misclassifications:** If a user says, "My boss is so stressed out", a basic system might mistakenly think the user is stressed. An LLM understands that the stress belongs to someone else (the boss) and ignores it.

2.3.2 Dealing with Ambiguous Responses

In real life, humans rarely speak in strict "Yes" or "No" answers. When asked if they are stressed, a user might reply with something unclear like, "I don't know", "Maybe", or "Just leave me alone". Instead of getting confused or forcing these answers into rigid "Stressed" or "Calm" buckets, a Large Language Model (LLM) can understand the nuance behind ambiguous human language.

Example: If the system asks, "Are you feeling stressed?" and the user replies, "Maybe, I am not sure", a traditional keyword system might completely fail to understand. An LLM, however, recognizes the implied meaning and can safely map this response to a specific category like AMBIGUOUS. This allows the system to interact with the user in a different way in order to understand, how the user is feeling in that moment.

2.3.3 Explainable AI (XAI)

Most AI systems operate like a black box; they make decisions, but they never tell the user how they made them. XAI changes this by having the AI explain its reasoning in simple, natural language [17].

For example, instead of a wearable device suddenly buzzing and asking, "Are you stressed?", an XAI system will explain the trigger: "I noticed your heart rate has been unusually high for the last 5 minutes. Are you feeling stressed?" By explaining the physical reason behind the question, the user understands the system isn't just randomly guessing, which increases their acceptance of the technology.

2.4 Real-Time Systems and Data Streams

In physiological monitoring, it is important to understand the difference between analyzing the data later and analyzing it right now. A real-time system processes incoming data and responds almost instantly. Unlike offline processing, where sensor data is saved to a file to be analyzed days or weeks later, a real-time system analyzes bodily signals the exact second they happen. This is crucial for a stress

detection framework because prompting the user or offering help is only useful if it happens while they are actually experiencing the stress.

- **Latency:** Latency is the lag or delay in the system. It is the total time it takes for a sensor to measure a physical reaction (like a heartbeat), send that data to the computer, run it through the AI models, and produce a final decision. In real-time systems, keeping latency very low (usually under a second) is critical so the system does not fall behind the live data.

To make a real-time system work smoothly without dropping data or crashing, specialized software frameworks are required to handle the communication between the sensors and the inference model.

- **LSL (Lab Streaming Layer):** When using multiple sensors, they often send data at completely different speeds. LSL is a software tool used in research to collect data from many different sensors over a network and perfectly synchronize them. It ensures that all the data shares the exact same clock, so the system knows exactly what the heart and the skin were doing at the exact same millisecond [18].
- **ROS 2 (Robot Operating System 2):** It is a software framework designed to help different parts of a complex system talk to each other quickly and reliably. In this stress detection framework, ROS 2 acts like a messenger. It takes the continuous stream of live sensor data and instantly delivers it to the machine learning models, ensuring the real-time system runs smoothly without losing any information [19].

3

State of the Art

This chapter evaluates the current state of the literature across three primary dimensions. First, it examines the foundational principles of Active Learning (AL) and the transition from offline sampling to real-time architectures. Second, it explores how these learning paradigms have been specifically adapted to physiological computing, addressing the unique challenges of signal noise and subject-wise variability in stress detection. Finally, the review considers the role of Human-Computer Interaction (HCI) and Large Language Models in transforming rigid classification outputs into empathetic, intent-aware dialogues. It also addresses the limitations in the previous work carried out in these fields.

3.1 Literature Review

This section evaluates the current literature across key technological and theoretical dimensions, establishing the foundational baseline for real-time stress monitoring and dialogue systems.

3.1.1 Foundations of Active Learning and Uncertainty Sampling

The core premise of this research relies on Active Learning, a paradigm where the algorithm selects the most informative data points for annotation rather than passively accepting a random training set. The foundational work on uncertainty sampling demonstrated that training classifiers on a selective subset of uncertain instances significantly reduces the labeling effort compared to random sampling [20]. This concept has been widely expanded in the domain of HITL machine learning. HITL systems can be categorized into model-oriented and data-oriented approaches. Furthermore, literature indicates that human feedback is most critical when model confidence is low or data distribution shifts [21].

Recent advancements have attempted to refine how uncertainty is measured. It has been argued that standard uncertainty sampling often selects noisy outliers rather than truly informative samples; consequently, weighing uncertainty against the density of the data distribution has been proposed to ensure robustness [22]. Similarly, in the domain of medical imaging, deep active learning has been shown to achieve state-of-the-art performance with only a fraction of the labeled data required by fully supervised models [23].

HITL systems emphasize that human feedback is most effective when used to refine a model's decision boundaries in real time. This approach has demonstrated significant viability in specialized fields like

neurotechnology, where Brain-Computer Interfaces (BCI) actively query users based on EEG uncertainty to improve classification accuracy [24]. Within this domain, the literature distinguishes between two primary scenarios for selecting informative data: Pool-Based Active Learning and Stream-Based Active Learning. Pool-Based Active Learning assumes the existence of a large, static collection of unlabeled data from which the model selects the most informative samples [25]. While effective for offline datasets such as medical image analysis, this approach is often ill-suited for real-time monitoring where data arrives sequentially. In contrast, Stream-Based Active Learning is designed for real-time environments where the model must decide instantaneously whether to query the user or discard the current data point [26]. These stream-based approaches are critical for wearable applications as they account for temporal shifts in data distribution, known as concept drift, and operate under strict latency constraints.

3.1.2 Active Learning in Stress Monitoring

While active learning is well-established in text and image domains, its application to high-dimensional, streaming physiological data presents unique challenges in timing and user burden. Traditional active learning often ignores the cost of a query, but in stress monitoring, an ill-timed question can itself become a stressor.

To address this, Active Reinforcement Learning (ARL) frameworks have been proposed to optimize the policy of when to ask. Empirical results demonstrate that such personalized, adaptive policies outperform static, fixed-interval query strategies in everyday settings by learning to balance model improvement against user interruption [27].

3.1.3 Conversational Interfaces and LLM Integration

While the detection logic is mathematical, the interaction layer is fundamentally linguistic. Traditional health chatbots often rely on static state machines, which frequently fail to handle complex context or user ambiguity. To address this, frameworks like Rasa have emerged as open-source conversational AI engines that separate Natural Language Understanding (NLU) from Dialogue Management [28]. Unlike rigid rule-based systems, these frameworks use machine learning to predict the next best action, allowing for the flexible, non-linear conversations required when a user is unsure of their own emotional state.

The interaction layer specifies how the system requests a label, which is as critical as the underlying algorithm. While traditional systems rely on rigid alerts, recent advancements in robotics suggest a hybrid approach. Integrating LLMs with curated NLU systems, such as Rasa, enables a hybrid architecture where the NLU handles precise intent detection and the LLM manages conversational flexibility and reflective dialogue. This curated NLU + LLM approach, further strengthened by Retrieval-Augmented Generation (RAG) from trusted knowledge bases, ensures the system remains reliable while appearing more empathetic and natural [29].

3.2 Limitations of Previous Work

The following table provides a comparative analysis of the existing literature against the core components of the proposed approach:

Paper	Real-time System	Physiological Signals	Active Learning	Dialogue Manager
Tazarv et al. (Active RL) [27]	✓	✓	✓	✗
Ulrich et al. (MISHA App) [30]	✗	✗	✗	✓
Battalio et al. (Sense2Stop) [31]	✓	✓	✗	✗
Ngoc et al. (DRAL) [32]	✓	✓	✓	✗
Browne et al. (Humanoid) [29]	✓	✗	✗	✓
Our Approach	✓	✓	✓	✓

Table 3.1: Comparison of research features across analyzed literature, ✓ represents the presence of the features, while ✗ suggests the absence of the features

In the pursuit of personalized health monitoring, Tazarv et al. [27] introduced a multilayered sensor-edge-cloud platform that captures signals such as photoplethysmogram (PPG) and accelerometer (ACC) in real time to monitor stress in everyday settings. This work implements context-aware active reinforcement learning to optimize the collection of user labels while minimizing the burden of ecological momentary assessments (EMAs), though it stops short of including a dialogue manager for conversational interaction. Similarly, Ngoc et al. [32] focused on high-accuracy stress recognition by proposing a Deep Reinforcement Active Learning (DRAL) framework that enhances long short-term memory (LSTM) models using physiological data from the Wearable Stress and Affect Detection (WESAD) dataset. While this approach demonstrates a robust real-time classification system driven by active learning, it does not incorporate a dialogue component for user engagement.

Shifting the focus toward interaction, Ulrich et al. [30] evaluated the MISHA app, a chatbot-delivered stress management coaching system that utilizes a text-based conversational agent to guide students through interventions. While this system features a dialogue manager to deliver coaching content, it does not operate as a real-time system or utilize physiological signals, relying instead on scheduled dialogues and self-reported data. Along a different path of intervention, Battalio et al. [31] developed the Sense2Stop trial, a real-time system using chest and wrist sensors to detect physiological stress to deliver just-in-time adaptive interventions. This system employs near real-time algorithms to determine the most effective timing for intervention; however, it lacks an active learning loop for model fine-tuning and a dialogue manager, as its interventions consist of static app-based exercises. Browne et al. [29] explored the use of a humanoid robot integrated with the Rasa framework and Large Language Models to conduct reflective dialogues aimed at positive behavioral change. This system highlights the potential of a highly capable real-time dialogue manager and NLU system, yet it does not currently integrate physiological signals for stress detection or active learning strategies to optimize the interaction loop.

Apart from the above features, it is noted that the vast majority of Active Learning research focuses

on pool-based scenarios [12]. There is a scarcity of robust stream-based frameworks specifically optimized for the noisy, multi-modal signal streams typical of wearable stress monitors. Wu et al. [21] notes that user engagement in HITL systems degrades if the interaction is perceived as burdensome or mechanical. Current systems fail to explain why they are confused; if a user does not understand the system's internal state, they may provide soft or inaccurate labels leading to lower user acceptance.

By combining these disparate elements, our approach aims to bridge the gap between real-time physiological sensing, active learning optimization, and dialogue management.

4

Methodology

This chapter outlines the methodological approach used to evaluate the real-time stress inference system. It details the data sources, hardware, and the experimental design used to test latency, model personalization, intent recognition, and user acceptance.

4.1 Evaluation Datasets and Sensors

To ensure robust evaluation across both physiological inference and semantic understanding, the system was tested using the following data sources and hardware:

- **WESAD Dataset:** Physiological data from 15 subjects in the WESAD dataset was utilized for latency and active learning experiments [33]. For this study, the baseline and stress conditions defined in the WESAD study protocol were utilized as the ground-truth labels. To test the system under real-world streaming constraints, the static CSV files were converted into a live data feed.
- **Emotibit Hardware:** To evaluate hardware streaming latencies under real-world conditions, live physiological streams were collected using an Emotibit wearable sensor [34].
- **SAD Dataset:** A sub-sampled benchmark from the Stress Annotated Dataset was used for binary stressor classification (1: is_stressor, 0: otherwise) [35].
- **Custom Conversational Dataset:** To evaluate the dialogue manager, a custom domain-specific dataset of 230 samples was generated using AI prompting. It was designed with challenging linguistic inputs, including temporal context shifts (e.g., past stressors), physical symptom reports, and ambiguous commands mapped across 8 intent categories: STRESS, CALM, ACCEPTANCE, REJECTION, AMBIGUOUS, PAUSE, RESUME, and UNKNOWN.

4.2 Setup

The system operates on a distributed Robot Operating System 2 (ROS 2) architecture. The sensory input layer is designed to process physiological streams from two distinct sources:

- **Simulated Environment:** A dedicated mock publisher node streams offline data from the WESAD dataset. This node manages sample rate synchronization and employs auto-jump logic to skip

unlabelled or non-relevant segments, ensuring a continuous stream of Blood Volume Pulse (BVP) at 64 Hz and EDA at 4 Hz.

- **Live Hardware:** An EmotiBit node connects to physical wearable sensors via the Lab Streaming Layer (LSL) protocol. It utilizes dedicated background threads to pull EDA (15 Hz) and specifically the Green PPG channel (25 Hz) in real-time; prioritized for its higher accuracy in extracting heart rate features, and publishes them to the ROS 2 network.

4.3 Experimental Procedures

This section outlines the step-by-step procedures used to conduct the evaluations for each research question.

4.3.1 Real-Time Streaming Latency Evaluation (RQ1)

To determine the end-to-end processing delay of the system under live conditions, a hardware latency test was conducted. A user wore the EmotiBit sensor continuously for approximately 15 minutes while the system ingested the live LSL data streams. Latency was measured programmatically using high-resolution performance counters (time.perf_counter()) initialized at the exact moment a 60-second data window was pulled from the buffer, and stopped the moment the inference prediction and entropy values were generated. These execution times were logged continuously to a local CSV file to evaluate whether the processing overhead exceeds practical real-time equivalence margins.

4.3.2 Active Learning and Model Personalization (RQ2 & RQ3)

The active learning and personalization framework was evaluated via a simulated continuous stream using the WESAD dataset. The physiological data was replayed to mimic a live wearable device in real-time. For every processing window, the model made a prediction. If the model was confused (H > 0.45), it triggered a query. As, this was a simulation, user feedback was simulated by having the system automatically fetch the actual ground-truth label from the WESAD dataset for that specific timestamp.

To avoid annoying the user, the evaluation system was given a budget of only two queries per continuous physiological episode. The first question allows the system to discover what a sudden change in physical signals means for that specific person. The second question acts as a safety net, allowing the system to solidify its confidence if the signals fluctuate. If the device asks a third or fourth time during that exact same event, it indicates the system has stubbornly failed to learn from the prior feedback. Therefore, any queries exceeding this two-question budget were tracked and flagged as redundant interruptions.

Upon receiving a label, the system built a personal model tailored to the user's unique physiological patterns. The evaluation script iteratively swept the blending weight (w) from 0.0 (100% generic baseline) to 1.0 (100% personal model) to empirically measure how increasing the reliance on personalized data affects overall accuracy and reduces redundant interruptions.

4.3.3 Intent-Aware Conversational Classification (RQ4)

The LLM intent classifier was evaluated against a standard keyword-based baseline. The custom 230-sample linguistic dataset was fed sequentially into the Qwen2.5:3B model via the engineered prompt system. The evaluation specifically tested whether the strict zero-temperature prompt constraints successfully prevented the LLM from inappropriately classifying physical symptoms and external complaints as real-time user stress, comparing the exact match accuracy against the rigid keyword rules.

4.3.4 User Acceptance and Natural Language Explainability (RQ5)

An online pilot user study was conducted ($N = 11$), to evaluate the user acceptance for XAI. The survey environment was built and hosted using SoSciSurvey [36]. A diverse cohort of participants was shown a video demonstration of the system in action and presented with a set of hypothetical interaction scenarios. Participants were asked to compare system queries that included physiological explanations against standard queries without explanations. Feedback and preferences were collected using a standard 5-point Likert scale ranging from Strongly Disagree to Strongly Agree and a direct comparative choice.

4.4 Experimental Design

This section outlines the independent and dependent variables, evaluation metrics, and statistical methods used to test the research hypotheses.

4.4.1 Variables and Evaluation Metrics

The evaluation measures system performance across four primary dimensions:

Table 4.1: Summary of experimental variables mapped to research questions.

Research Question	Independent Variables	Dependent Variables
RQ1	Streaming condition (Pre-Processed vs. Simulated vs. Live)	Latency per window (ms)
RQ2 & RQ3	Model type (Static vs. Adaptive)	Accuracy, F1-Score, Query count, False Positive Rate
RQ4	Model (Keyword vs. LLM)	Accuracy, F1-Score, Exclusive corrects
RQ5	Explanation state (With vs. Without)	Likert scores (1-5), Option preference (%)

4.4.2 Statistical Analysis Framework

Hypotheses are tested using the following statistical procedures:

- **RQ1:** The Shapiro-Wilk test is first applied to assess data normality, determining whether the latency distributions follow a standard Gaussian curve [37]. Because system latencies often violate normality assumptions, a non-parametric Bootstrap Confidence Interval for equivalence is utilized [38]. Bootstrapping involves repeatedly resampling the data with replacement to estimate the true distribution of the mean difference. To establish equivalence at a standard significance level of $\alpha = 0.05$, a 90% confidence interval is constructed for the latency differences between conditions. If this entire 90% interval falls strictly within the predefined practical equivalence margin of $1,000$ ms with a Bonferroni correction, the processing delay is considered practically insignificant.

- **RQ2 & RQ3:** The transition from a static baseline model to an adaptive model is evaluated via a simulated continuous stream. The experiment tests maximum blend weights (w) ranging from 0.1 to 1.0. Evaluation criteria include standard classification metrics (Accuracy, F1-Score, Confusion Matrix) alongside query frequency metrics to measure user burden.

- **RQ4:** Performance is measured via Accuracy and F1-Score. McNemar's test is applied to evaluate whether differences in classification correctness between the keyword baseline and the Qwen2.5:3B model are statistically significant ($\alpha = 0.05$) [39]. McNemar's test is specifically designed for paired nominal data; in this context, it evaluates the contingency table of both models predicting the exact same dataset to definitively prove if one model disproportionately succeeds where the other fails.

- **RQ5:** Explanations are generated using feature attribution from the Random Forest. SHAP (SHapley Additive exPlanations) offers a robust way to interpret machine learning results by isolating the contribution of every feature. This method assigns a unique importance value to each input for every prediction made, ensuring that the resulting explanations are both fair to the data and consistent across different scenarios [40]. SHAP is utilized to extract the contribution of individual physiological features to the model's predictions. A TreeExplainer model evaluates samples extracted from 60-second processing windows. The analysis plots the SHAP values against the 16 extracted features to interpret the classification of calm and stress states. A user study evaluates participant perception of uncertainty-triggered queries equipped with natural language explanations.

5

Solution

This chapter details the core components of the real-time stress inference framework. It outlines the conceptual system architecture, the algorithms driving signal processing, active learning and semantic dialogue manager, and the implementation specifics that ensure real-time performance.

5.1 System Architecture

The system operates as a closed-loop, HITL active learning framework. The individual wearing the sensors acts as both the primary physiological data source and the system supervisor. As illustrated in Figure 5.1, the architecture is divided into four continuous stages:

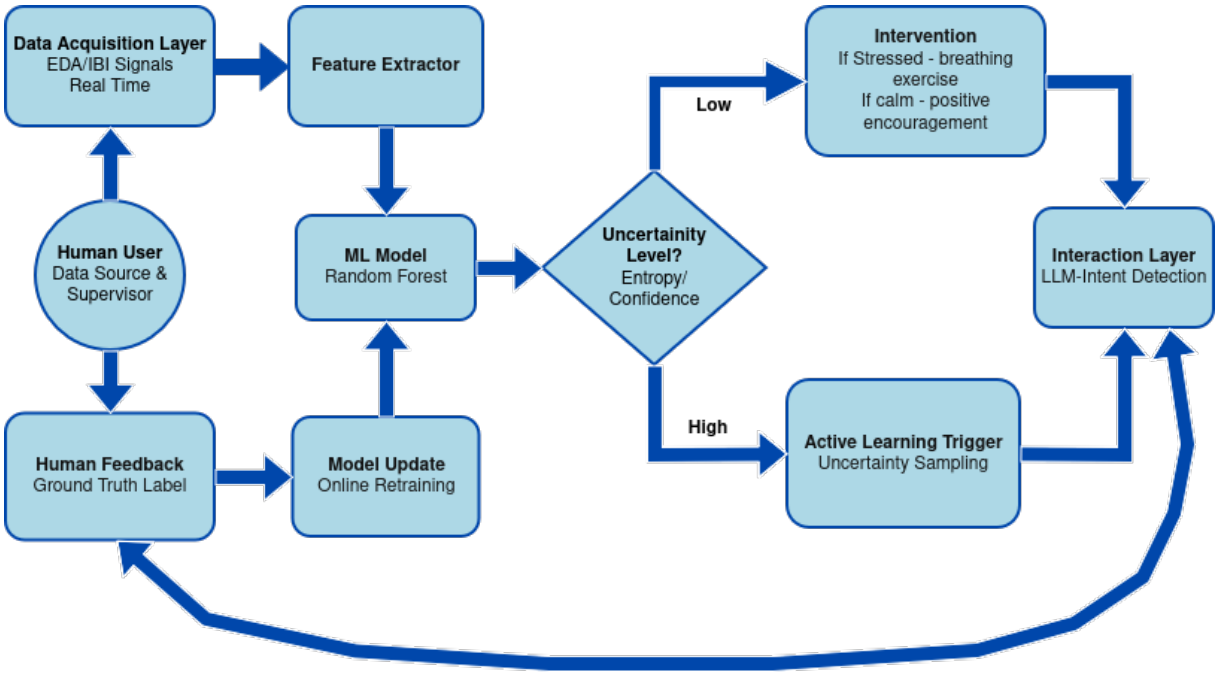


Figure 5.1: Human-in-the-Loop (HITL) active learning system architecture for real-time stress monitoring and intervention.

- **Data Acquisition and Processing:** The Data Acquisition Layer collects real-time EDA and Inter-Beat Interval (IBI) derived from BVP signals from the user. These raw streams are passed to the Feature Extractor, which isolates relevant statistical markers.
- **Core Inference and Decision Routing:** The extracted features are fed into the machine learning model, utilizing a Random Forest classifier to predict the user's stress state. The selection of underlying classification algorithms is critical for real-time physiological systems. Ensemble methods, particularly Random Forest, are widely preferred over more complex deep learning architectures for real-time streaming because they provide high inference speeds, and are lightweight [41]. A decision gate then evaluates the uncertainty level (entropy) of this prediction to determine the execution pathway.
- **Execution Pathways:**
  - **Low Uncertainty:** Confident predictions bypass retraining and route directly to the Intervention module, which triggers a breathing exercise if the user is stressed, or provides positive encouragement if the user is calm.
  - **High Uncertainty:** Unconfident predictions route to the Active Learning Trigger, utilizing uncertainty sampling to flag the data point for manual review.
- **Interaction and Continuous Learning:** Both execution pathways converge at the Interaction Layer, where a LLM handles Intent Detection and converses with the user. Through this interface, the system acquires Human Feedback (a Ground Truth Label). This label is sent to the Model Update module for online retraining, allowing the machine learning model to continuously adapt to the user's unique baseline.

5.2 Proposed Algorithm

The proposed algorithmic pipeline is designed to transform raw physiological signals into actionable, personalized stress interventions. This continuous process is divided into three sequential stages: feature extraction from sensory streams, uncertainty-aware stress classification, and semantic dialogue management. The specific methodologies for each stage are detailed below.

5.2.1 Signal Processing and Feature Extraction

The sensory pipeline handles raw data ingestion from real-world hardware (such as the EmotiBit sensor capturing EDA at 15 Hz and PPG at 25 Hz) and aligns it with the operational requirements of the offline-trained WESAD baseline model (4 Hz EDA and 64 Hz BVP). To manage continuous streaming without memory bloat, the system implements dual time-windowed buffers using double-ended queues (deque). The buffers maintain fixed maximum capacities scaled directly by the input sampling rates and window duration. Processing operates on a 60-second window. The system utilizes signal processing libraries to clean the incoming signals and compute 16 distinct features per window:

- **EDA Features:** Mean, standard deviation, minimum, maximum, range, skewness, kurtosis, Tonic standard deviation, and Skin Conductance Response (SCR) peak amplitude mean and count.
- **IBI Features:** IBI mean, standard deviation, median, minimum, maximum, and range.

5.2.2 Uncertainty-Aware Active Learning

Initial stress inference relies on a static Random Forest classifier trained on population-level data. To compute predictive uncertainty, the system calculates the Shannon entropy on the model's output probabilities:

$$H = -\sum_{i} p_i \log_2 p_i$$

If the entropy exceeds a defined threshold ($H > 0.45$) for a sustained period, the Active Learning Trigger prompts the user for a self-reported label [42].

Upon receiving explicit user feedback, the system stores the corresponding feature vector and its associated label into a persistent user dataset. Because physiological data arrives continuously, this framework specifically implements stream-based active learning. Rather than pulling from a static offline pool, the model evaluates live data windows sequentially, making instantaneous decisions on whether to query the user or proceed with automated intervention [26]. Once a minimum threshold of labeled samples is reached ($n \geq 2$), the system initiates an online model update. The feature matrix is normalized using a standard feature scaler StandardScaler, and a personalized RandomForestClassifier is trained using 50 estimators and balanced class weights is build to account for potential skewness between calm and stress occurrences. To prevent blocking the primary ROS 2 node, this retraining routine executes asynchronously inside a background thread.

Subsequent inferences utilize a blended probability ($P_{blended}$) combining the base population model (P_{base}) and the personal model ($P_{personal}$):

$$P_{blended} = (1 - w)P_{base} + wP_{personal}$$

The dynamic blend weight (w) scales with the number of accumulated personal labels (n), up to a defined maximum weight (w_{max}):

$$w = \min\left(w_{max}, \frac{n}{N_{sat}}\right)$$

where N_{sat} represents the label saturation threshold.

As the user continues to use the system over time, the number of provided labels (n) will eventually grow very large. Without the min function capping the result, the weight calculation would eventually exceed 1.0, which would result in invalid probability mathematics and crash the system. Furthermore, this blending probability approach is used instead of replacing the base model to solve two major challenges in continuous learning:

- **Preventing Catastrophic Forgetting:** If the system completely replaced the population model with a personal model trained on just a few new labels, it would instantly forget all the general rules

of physiological stress. Blending them allows the system to retain its broad foundational knowledge while gradually mixing in the user's unique patterns. [43].

- **Handling Noisy Labels:** In real-world settings, users make mistakes (e.g., clicking stressed when they are actually just physically tired from walking). If the system gave the personal model full control after only two or three labels, a single wrong answer would severely confuse the system. By making the weight (w) to scale up slowly over time, the system protects itself. It relies safely on the general base model at first, and only trusts the personal model more as it gathers enough data to be statistically confident [44].

5.2.3 State Machine and Intervention Logic

To govern system behavior, prevent user alert fatigue, and manage conversational pacing, the framework implements a deterministic finite state machine. The system tracks state transitions and user responsiveness using the temporal constraints and timeout parameters detailed in Table 5.1.

Table 5.1: State machine parameters and temporal timeout configurations.

Parameter	**Description**
Uncertainty Streak	Duration of sustained high entropy ($H > 0.45$) before triggering an active learning query.
Calm Streak	Cooldown period following a calm intervention to prevent over-prompting.
Stress Streak	Cooldown period following a stress intervention or active learning query.
Reply	Maximum wait time for user feedback before sending a check-in reminder.
Reminder	Time elapsed after the reminder before abandoning the active query window.
Breathing Cycle	Duration of a single automated breathing exercise cycle.
Total breathing cycles	Total number of cycles in an automated breathing exercise.

The temporal timeouts defined in 5.1 are grounded in both user experience design and human biological response times. Because the system evaluates physiological data and generates predictions continuously, intervention logic would result in repeated alerts. Research indicates that users become significantly less likely to accept system interventions as the frequency of repeated alerts increases, leading to alert overrides and alert fatigue [45]. Therefore, the Uncertainty Streak and Calm/Stress Streaks mitigate alert fatigue by suppressing redundant prompting.

5.2.4 Semantic Dialogue Management

At the Interaction Layer, user text responses are analyzed by a localized LLM intent classifier (Qwen2.5:3b) configured with zero temperature for deterministic text classification. Although the underlying physiological Random Forest model evaluates user states in a strictly binary format (Calm or Stress), mapping human dialogue requires a higher-dimensional approach. If the dialogue manager relied on binary intent mapping, any user response that is not explicitly positive (e.g., "I am not sure", "Leave me alone", "I have a headache") would be forced into a rigid stress or non-stress category, leading to inappropriate physiological labeling and unsafe interventions. To prevent false interventions and handle complex linguistic edge cases, the system enforces a strict classification via system prompts that categorize user text into one of eight distinct intent tags, as outlined in Table 5.2.

Table 5.2: Semantic intent classification tags and mapping rules.

Intent Tag	**Description**
STRESS	User explicitly states internal stress or anxiety in the present tense (e.g., "I am stressed", "I feel anxious").
CALM	User explicitly states they are currently relaxed, feeling good, or doing well.
ACCEPTANCE	User explicitly agrees to a breathing exercise or intervention (e.g., "yes", "ok", "sure").
REJECTION	User explicitly declines an offered intervention (e.g., "no thanks").
AMBIGUOUS	Expresses uncertainty or mixed states (e.g., "maybe", "I don't know", "not sure").
PAUSE	User requests to stop, pause, or leave monitoring.
RESUME	User requests to restart or resume monitoring.
UNKNOWN	Default fallback category; strictly mandated for past-tense statements, external complaints (e.g., "traffic", "work"), etc.

Based on the resolved intent, the dialogue manager governs execution paths: ACCEPTANCE initiates an automated breathing exercise protocol; REJECTION responses prevent the system from trapping the user in repetitive follow-up loops; AMBIGUOUS triggers clarifying questions to better assess the user's current state; UNKNOWN handles daily conversations, past-tense entries, physical symptoms, or external complaints with general supportive acknowledgment; and PAUSE/RESUME manage system monitoring controls. The state machine enforces strict cooldown limits following interventions to minimize user burden.

5.2.5 Explainable AI (XAI) and Model Interpretability

To ensure transparency and to bridge the gap between machine learning predictions and user understanding, an XAI layer utilizing SHAP is useful. SHAP is a game-theoretic approach to explain the output of any machine learning model [46]. It calculates the marginal contribution of each physiological feature to the final prediction, ensuring a fair and consistent distribution of credit among the input variables. Unlike standard feature importance metrics that only provide global model insights, SHAP offers mathematically robust local interpretability, allowing the system to isolate exactly which physiological spikes caused a specific stress alert at a precise moment in time. This interpretability framework is structured around both global model behavior and local real-time explanations:

- **Global Interpretability (Overall Model Behavior):** The overarching decision-making logic of the Random Forest classifier is evaluated using global feature importance rankings. As illustrated in Figure 5.2, EDA features heavily dominate the model, with skin conductance variability markers (eda_std, eda_range, and eda_max) acting as the strongest primary indicators of stress within the WESAD dataset, while IBI features play a secondary, supportive role.

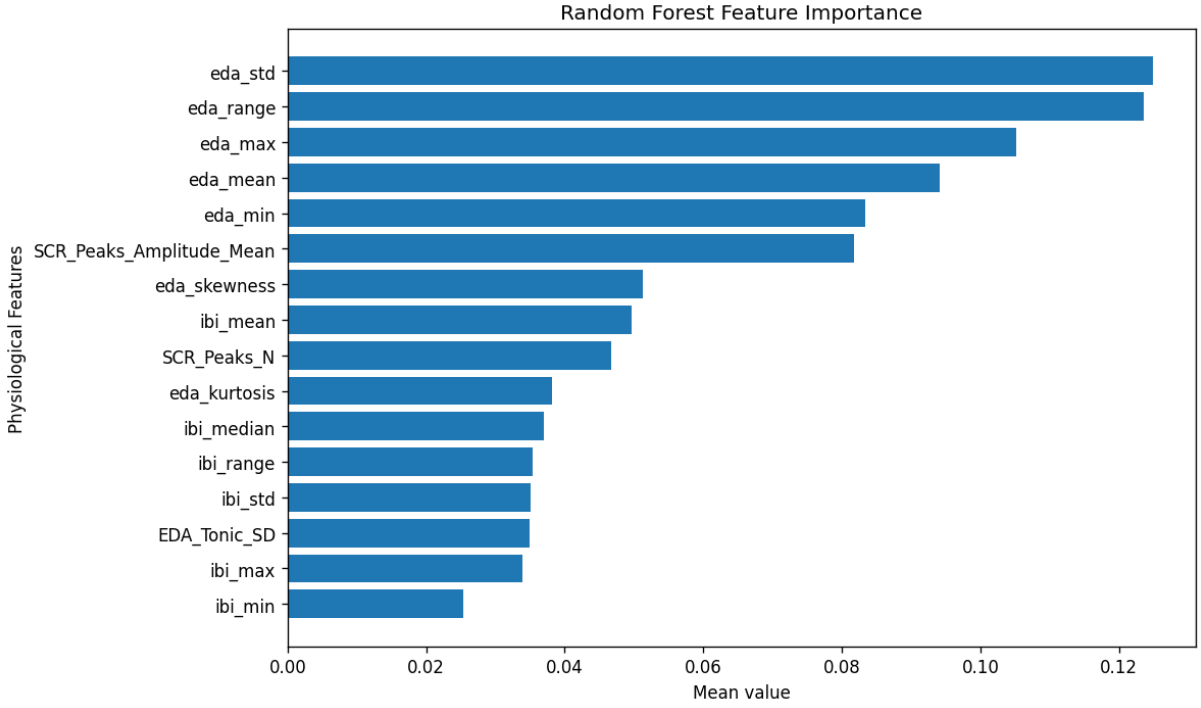


Figure 5.2: Global Feature Importance across the WESAD dataset.

Class-specific mean SHAP evaluations further confirm model balance (Figure 5.3), demonstrating that the classifier relies on the exact same core physiological logic to detect a calm state as it does to detect a stressed state, rather than utilizing disjointed feature sets.

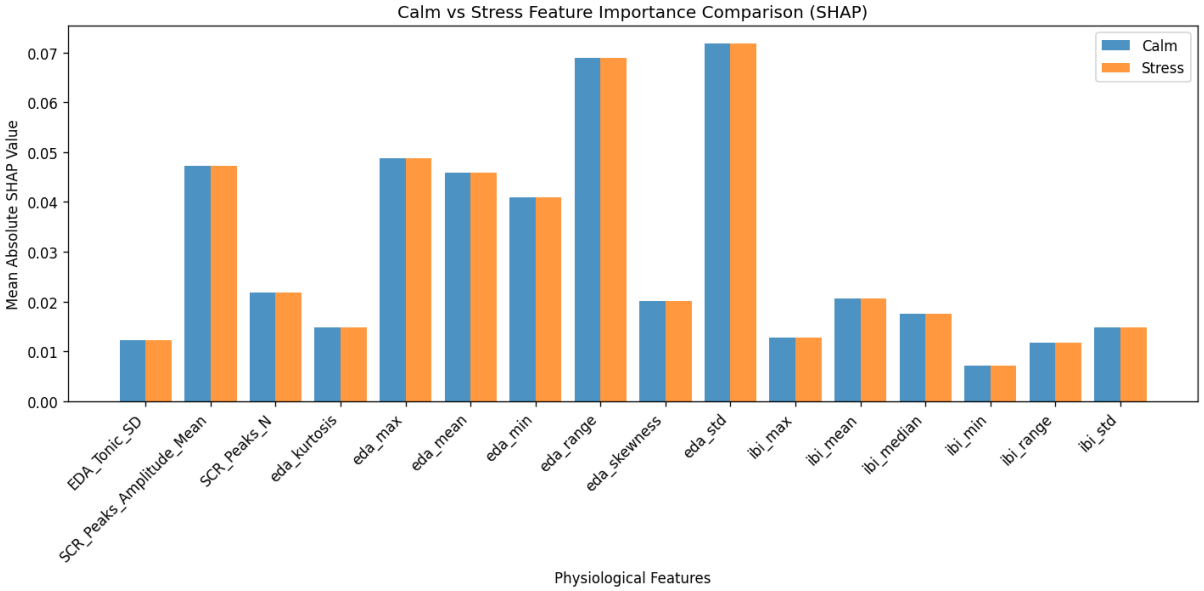
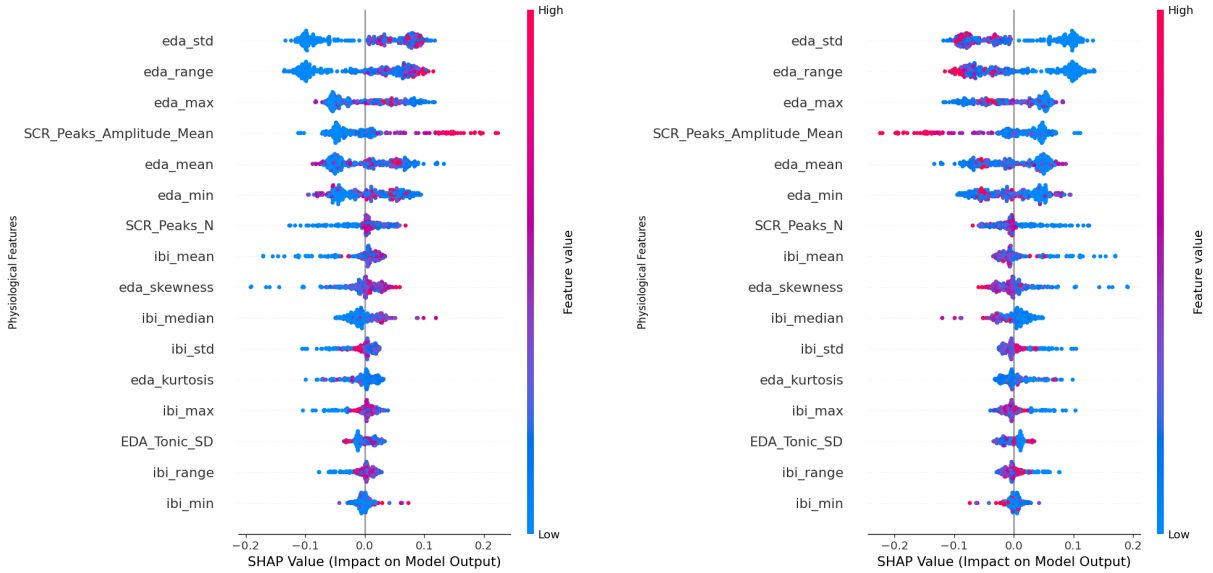


Figure 5.3: Class-Specific Mean SHAP Bar Chart showing mean absolute SHAP values split by class.

- **SHAP Summary Dot Plots:** Feature importance and directional impact are mapped via class-specific summary dot plots for Stress (Figure 5.4a) and Calm (Figure 5.4b). Features are ranked vertically by importance, while horizontal SHAP values dictate class probability shifts (positive values increase probability, negative values decrease it). Raw physiological magnitudes are color-coded to visualize feature thresholds.

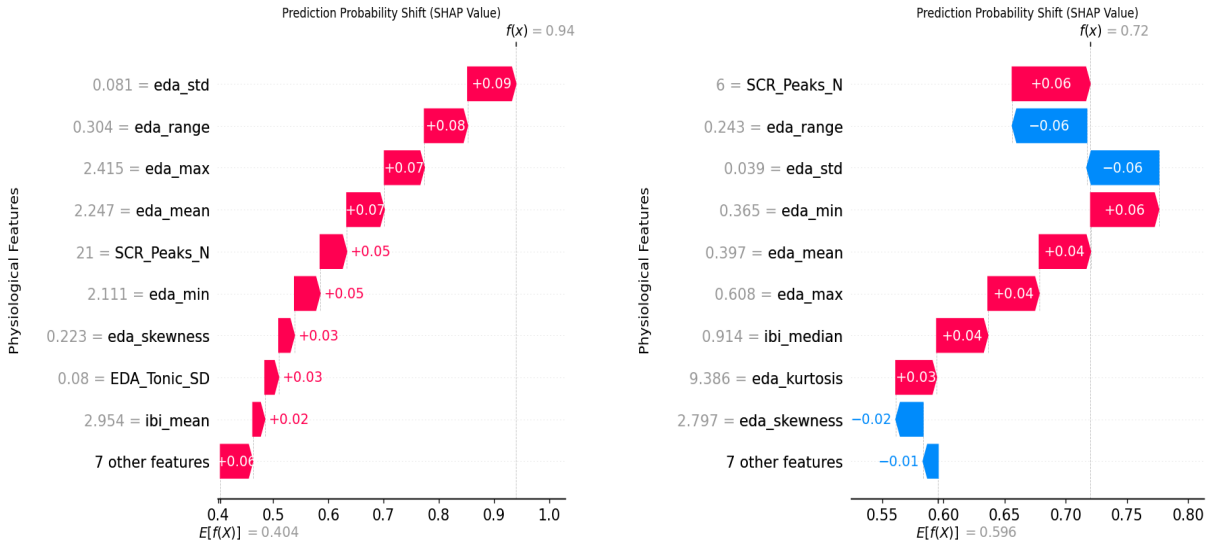


(a) Dot Plot for Stress

(b) Dot Plot for Calm

Figure 5.4: SHAP summary dot plot illustrating feature importance, directionality, and raw physiological values for predicting Stress and Calm. The vertical axis lists features ordered by overall importance from top to bottom. The horizontal axis represents the SHAP value, where points to the right of zero increase the model's prediction and points to the left decrease it. Each dot represents a single data instance, with color indicating the feature's relative value (red for high values, blue for low values).

- **Local Interpretability (Single-Moment Explanations):** To decode individual predictions at precise temporal instances, local waterfall plots isolate single inference windows for stress events (Figure 5.5a) and calm events (Figure 5.5b). Starting from a baseline expected probability $E[f(x)]$ to a final prediction probability $f(x)$, directional arrows quantify real-time feature contributions. Red arrows highlight physiological spikes increasing the target state probability, while blue arrows represent counteracting stabilizing features.



(a) Local Explanation for a Stress Event

(b) Local Explanation for a Calm Event

Figure 5.5: Local Explanation Waterfall Plot breaking down a single moment in time. The horizontal axis indicates the model prediction output. The vertical axis lists the input features along with their specific values for this instance, ordered by impact magnitude from top to bottom. Red and blue bars denote positive and negative feature contributions, respectively.

- **Natural Language Translation of Explanations:** Rather than issuing opaque or generic notifications, the XAI framework translates complex waterfall outputs into transparent, contextual dialogue. For instance, parsing a stress event (such as Figure 5.5a) where red arrows indicate sharp rises, the system replaces clinical alerts with descriptive dialogue, translating raw attributions directly into conversational prompts: "Your skin conductance variability and range have spiked significantly. Are you feeling stressed?"

5.3 Implementation Details

The framework is implemented in Python within a distributed ROS 2 environment. The software design leverages several key technical libraries, communication paradigms, and mechanisms to ensure modularity, safety, and strict real-time performance:

- **ROS 2 Node and Communication Infrastructure:** The core logic is encapsulated within the inference node. Communication across the system is managed through dedicated ROS 2 topics and message types. The node subscribes to real-time sensor streams (eda_raw, bvp_raw), user input channels (user_feedback, ground_truth), and broadcasts dynamic system updates, including dialogue prompts, state changes, and visual statuses. Furthermore, system execution is governed by ROS 2 periodic timers handling sliding-window processing, breathing exercise watchdogs, and reply-window timeouts.

- **Signal Processing, Resampling, and Feature Extraction Stack:** Signal cleaning, artifact removal, and physiological feature extraction rely heavily on the NeuroKit2 library. A critical technical challenge in real-time inference is hardware mismatch; the live EmotiBit sensor streams EDA at 15 Hz and PPG at 25 Hz, whereas the baseline model expects WESAD formatted data at 4 Hz and 64 Hz, respectively. To resolve this, raw signals are first buffered and cleaned at their native input frequencies (nk.eda_clean and nk.ppg_clean), and then dynamically resampled to the model's required frequencies during interval-related metric extraction and peak finding algorithms (nk.ppg_findpeaks) to compute IBI markers.

- **Software Stack and Model Persistence:** Pre-trained population baseline models and feature normalizers are instantiated using scikit-learn and serialized for real-time deployment via joblib (specifically loading the saved scaler and model weights). The semantic dialogue manager integrates locally with Ollama using the Qwen2.5:3b model, fixing the inference temperature parameter to 0.0 to obtain deterministic text classification outputs.

- **Real-Time Concurrency and LSL Streaming:** To maintain low-latency performance and prevent processing stalls, concurrency is explicitly managed across the architecture. The live hardware data acquisition node utilizes dedicated background threads (threading.Thread) to isolate blocking LSL pull_sample requests. This design ensures the continuous, simultaneous ingestion of both the EDA stream and the Green PPG channel, prioritized for its superior signal-to-noise ratio in heart rate extraction; without dropping frames or blocking the ROS 2 node graph [47].

- **Asynchronous Active Learning, Thread Safety, and Logging:** Online model retraining is computationally expensive and can introduce latency spikes that disrupt real-time control loops. The Active Learning Manager resolves this by utilizing background threads paired with thread synchronization primitives (threading.Lock). This isolates personalization updates so that user-triggered feedback and model retraining execute asynchronously. Additionally, system metrics, entropy values, query history, and active learning labels are continuously serialized and logged to local CSV files within an active learning directory for downstream evaluation.

6

Evaluation and Results

6.1 Experiment Description

This chapter evaluates the performance, personalization capabilities, intent understanding, and user acceptance of the proposed real-time stress detection and active learning framework.

1. **Real-Time Latency Analysis:** Evaluates system execution latency across three streaming conditions: pre-processed static data, simulated real-time data streams, and live sensor input. This tests whether transitioning to live, noisy physiological streams introduces latency overhead that impairs real-time operation.

2. **Active Learning and Personalization Performance:** Measures the impact of an adaptive active learning mechanism on reducing redundant interruptions and improving stress detection performance over time. Results are benchmarked across a range of adaptation weight parameters ($w \in [0.1, 1.0]$) against a static, non-adaptive baseline model.

3. **Intent-Aware Dialogue Mapping:** Compares an LLM-based intent classifier (Qwen2.5:3B) against a standard keyword-based rule system. The models are evaluated on a custom domain-specific dataset containing complex linguistic edge cases and a subset of the public Stress Annotated Dataset (SAD).

4. **User Acceptance and Natural Language Explainability:** Evaluates user preference and perception of uncertainty-triggered queries equipped with natural language explanations compared to unexplained queries through a user study ($N = 11$).

6.2 Experimental Setup

This section details the empirical environment, hardware infrastructure, benchmark datasets, baseline models, and evaluation methodologies used across all experiments.

6.2.1 Models

- **Static vs. Adaptive Model:** A static model using fixed prediction entropy thresholds was compared against an adaptive active learning model that updates model weights (w) based on user feedback during prolonged physiological states.
- **Keyword-Based vs. LLM Intent Classifier:** A keyword matching pipeline was benchmarked against a LLM model (Qwen2.5:3B), it was evaluated using accuracy, F1-score, exclusive correct classifications, and McNemar's test for statistical significance.

6.2.2 User Study Setup

A cohort of $N = 11$ participants was surveyed (demographics: 63.6% male, 36.4% female; 54.5% aged 18-24, 36.4% aged 25-34, 9.1% aged 35-44; 72.7% prior wearable device experience). Participants rated explainability features on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree) across four criteria (QXAI_01 to QXAI_04) and indicated explicit preference between explained and unexplained system queries (QXAI) [48].

- **QXAI_01:** This explanation shows me how reliable the system is.
- **QXAI_02:** This explanation helps me know when I should trust and not trust the system.
- **QXAI_03:** This explanation of how the system works has sufficient detail.
- **QXAI_04:** This explanation shows me how accurate the system is.

System preference (QXAI) was evaluated through a direct comparative choice:

- **Option A:** System queries without explanations.
- **Option B:** System queries accompanied by natural language explanations of physiological prediction triggers.

6.3 Results

This section presents the quantitative findings and statistical analyses corresponding to each research question. The evaluation progresses from system-level hardware performance and active learning adaptation to dialogue mapping accuracy and user study feedback. Formal hypothesis testing (H_0/H_1) is reported for each experimental condition to evaluate statistical significance.

6.3.1 Real-Time Streaming Latency Evaluation

RQ1: Can a real-time streaming architecture maintain comparable stress detection performance when transitioning from streams of pre-processed data to raw, noisy physiological signals?

- **Null Hypothesis ($H_{0,1}$):** There is a significant difference in stress detection performance when the input source switches from pre-processed dataset streams to raw, real-time sensor feeds.
- **Alternative Hypothesis ($H_{1,1}$):** There is no significant difference in model performance when the input source switches from pre-processed dataset streams to raw, real-time sensor feeds.

To address RQ1, system execution latency was measured across three input streaming conditions to determine the processing overhead introduced by live hardware streaming. Table 6.1 presents the statistical distribution of response times.

Table 6.1: System processing latency (ms) across data streaming conditions.

Statistic	Pre-processed Dataset	Simulated Stream	Live Hardware Stream
Mean	6.14	53.87	295.50
Median	5.83	56.85	293.77
Std Dev	1.06	19.90	45.48
Min	5.42	21.23	179.67
Max	10.83	135.46	524.68

Using a predefined significance level of $\alpha = 0.05$, Shapiro-Wilk tests confirmed non-normal distributions across all conditions: Pre-processed ($p = 4.64e^{-12}$), Simulated Stream ($p = 0.0003$), and Live Hardware ($p = 0.0002$).

A non-parametric bootstrapping test yielded $p < 0.0001$. The 90% confidence intervals for pairwise differences in execution time were:

- **Pre-processed vs. Simulated Stream:** $[-52.06 \text{ ms}, -43.54 \text{ ms}]$
- **Pre-processed vs. Live Hardware Stream:** $[-300.50 \text{ ms}, -278.40 \text{ ms}]$
- **Simulated Stream vs. Live Hardware Stream:** $[-253.34 \text{ ms}, -229.72 \text{ ms}]$

Although moving to live hardware increases processing latency from 6.14 ms to 295.50 ms due to real-time sampling and signal filtering, the maximum observed latency (524.68 ms) remains far below the pre-defined practical equivalence margin (δ = 1,000 ms / 1 seconds). As the reviewed literature lacks a universal standard for acceptable delays in real-time stress detection, this 1,000 ms limit was established as a practical engineering constraint. This threshold aligns with similar physiological computing frameworks, ensuring the system responds fast enough for seamless human-in-the-loop interaction. Evaluating system latency against application-specific clinical benchmarks is a standard validation practice in physiological

computing [49]. Therefore, an end-to-end processing delay of roughly 1 second is functionally instantaneous for stress monitoring [50]. Because the latency falls safely within the required equivalence margin, $H_{0,1}$ is rejected, confirming that the real-time streaming framework operates comfortably within acceptable system constraints.

A post-hoc statistical power analysis was conducted on the equivalence test. Using the predefined equivalence margin of 1,000 ms and a sample size of $N = 60$ processing windows per condition, the calculated statistical power for all three comparisons (Pre-processed vs. Simulated, Pre-processed vs. Live Hardware, and Simulated vs. Live Hardware) was 100%. This guarantees that the sample size was sufficient, proving that the system operates safely and reliably in real-time.

6.3.2 Active Learning and Model Personalization

RQ2: Can an active learning framework reduce the frequency of redundant interruption during prolonged physiological episodes compared to a static model?

- **Null Hypothesis ($H_{0,2}$):** There is no statistically significant difference in the redundant interruption rate between a static model with fixed uncertainty rules and an adaptive active learning model.
- **Alternative Hypothesis ($H_{1,2}$):** The adaptive active learning model significantly reduces the redundant interruption rate compared to the static model.

RQ3: To what extent active learning with uncertainty sampling improve the personalization accuracy of stress detection compared to a static, subject-independent model?

- **Null Hypothesis ($H_{0,3}$):** There is no significant difference in stress detection accuracy between a static, subject-independent model and an active learning model with uncertainty sampling.
- **Alternative Hypothesis ($H_{1,3}$):** An active learning model with uncertainty sampling achieves significantly higher personalization accuracy and lower false-positive rates compared to a static, subject-independent baseline by progressively updating on user-specific data.

Addressing RQ2 and RQ3, the performance of the adaptive active learning framework was evaluated against a static model across varying adaptation weight parameters ($w \in [0.1, 1.0]$). Table 6.2 summarizes the total query count and redundant interruptions across the WESAD dataset. Table 6.3 summarizes detection accuracy and F1-score.

Table 6.2: Evaluation of query frequency and redundancy across model adaptation weights.

	Static Model		**Adaptive Model**	
Weight (w)	**Queries**	**Redundancy**	**Queries**	**Redundancy**
0.1	19.1	15.7	23.4	20.0
0.2	19.1	15.7	23.8	20.4
0.3	19.1	15.7	22.9	19.5
0.4	19.1	15.7	22.9	19.5
0.5	19.1	15.7	22.9	19.5
0.6	19.1	15.7	22.9	19.5
0.7	19.1	15.7	22.9	19.5
0.8	19.1	15.7	20.5	17.1
0.9	19.1	15.7	18.2	14.8
1.0	19.1	15.7	18.1	14.7

Evaluating query redundancy reveals that at lower adaptation weights ($w \leq 0.8$), the adaptive model yields higher redundant interruptions (17.1 to 20.4) compared to the static baseline (15.7). However, as the personalized model is given higher weight ($w \geq 0.9$), redundancy of the query decreases significantly, dropping to 14.7. Thus, $H_{0,2}$ can be rejected, as these results demonstrate that the redundant interruption frequency decreases relative to the static model when the personalized model exerts a dominant influence over system decisions.

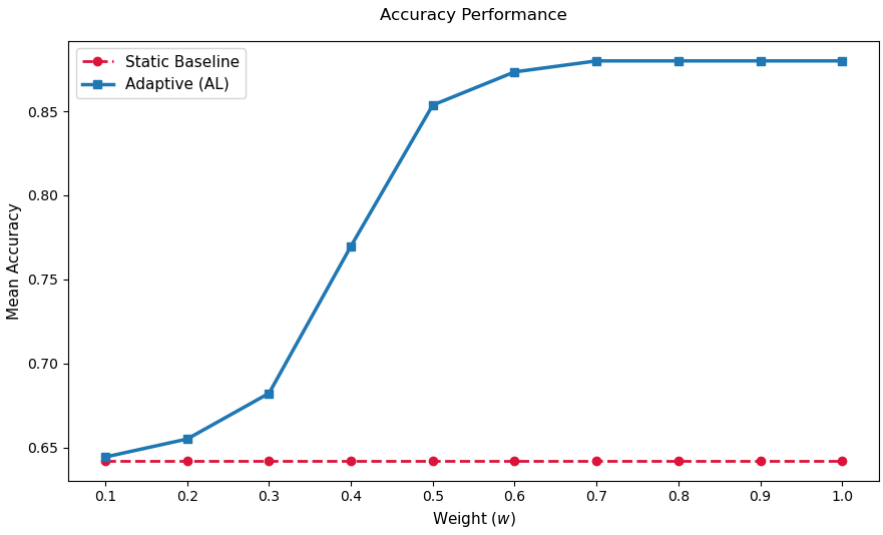


Figure 6.1: Impact of adaptation weight (w) on overall detection accuracy.

Table 6.3: Classification performance comparison across adaptation weights.

	Static Model		Adaptive Model	
Weight (w)	**Accuracy**	**F1-Score**	**Accuracy**	**F1-Score**
0.1	64.2%	1.7%	64.4%	2.6%
0.2	64.2%	1.7%	65.5%	7.4%
0.3	64.2%	1.7%	68.2%	17.9%
0.4	64.2%	1.7%	77.0%	47.9%
0.5	64.2%	1.7%	85.4%	73.0%
0.6	64.2%	1.7%	87.3%	77.6%
0.7	64.2%	1.7%	88.0%	78.9%
0.8	64.2%	1.7%	88.0%	78.9%
0.9	64.2%	1.7%	88.0%	78.9%
1.0	64.2%	1.7%	88.0%	78.9%

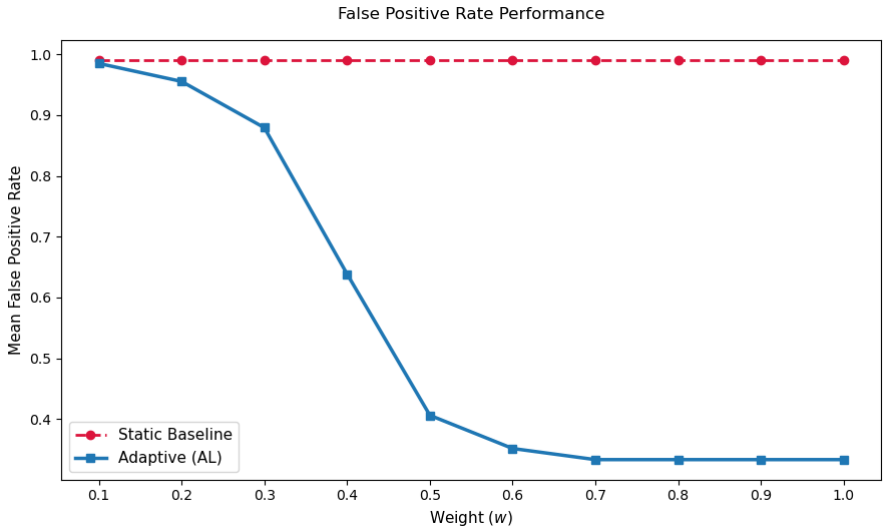


Figure 6.2: Mean False Positive Rate (FPR) across adaptation weights (w) comparing the Static Baseline against the Adaptive Active Learning model, with Calm designated as the positive class (1) and Stress as the negative class (0).

Defining Calm as the positive class (1) and Stress as the negative class (0) allows the False Positive Rate to measure the proportion of actual stress episodes incorrectly classified as calm (i.e., missed stress detection). As shown in Figure 6.2, the reduction ($\sim$ 99.0% to $\sim$ 20%) confirms that prioritizing personalized feedback allows the framework to correctly identify true stress states rather than defaulting to non-stress predictions.

Increasing the adaptation weight significantly improves stress detection performance. As personalization weight increases, overall accuracy rises from 64.2% to 88.0% ($w \geq 0.7$), while the F1-score increases from 1.7% to 78.9%. Both classification accuracy and F1-score plateau at $w = 0.7$ (Figure 6.1). This saturation occurs due to dataset limits in WESAD, where the finite number of labeled subject samples bounds the additional gains achievable through further personalization weight increases. These findings reject $H_{0,3}$.

6.3.3 Intent-Aware Conversational Classification

RQ4: Does the use of intent-aware LLM queries improve the correctness of mapping ambiguous user responses to ground-truth stress labels compared to keyword-based rule systems?

- **Null Hypothesis ($H_{0,4}$):** There is no statistically significant difference in mapping accuracy between the LLM-based intent classifier and the keyword-based rule system.
- **Alternative Hypothesis ($H_{1,4}$):** The LLM-based intent classifier achieves significantly higher mapping accuracy than the keyword-based rule system.

To answer RQ4, the LLM-based intent classifier (Qwen2.5:3B) was evaluated against the keyword rule baseline across both the custom domain-specific dataset and the public SAD benchmark dataset. Table 6.4 presents the comparative evaluation metrics.

Table 6.4: Intent mapping performance across evaluation datasets.

	Custom Dataset ($n = 230$)		**SAD Benchmark** ($n = 230$)	
Metric	**Keyword**	**LLM**	**Keyword**	**LLM**
Overall Accuracy	48.07%	80.69%	35.47%	82.91%
Macro F1-Score	0.439	0.723	0.314	0.628
Exclusive Correct Classifications	18	94	12	123
McNemar's Test (p-value)	$1.34e^{-13}$		$2.35e^{-24}$	

As shown in Table 6.4, the LLM model achieved slightly higher overall accuracy on the binary SAD benchmark (82.91%) compared to the custom 8-class dataset (80.69%). While binary classification often yields higher baseline accuracy by artificially simplifying the problem space, this reductionism is unsafe in real-world conversational interfaces. Human responses are rarely a strict "stressed" or "not stressed". A binary framework forces nuanced responses, such as "I don't know" or "leave me alone"; into rigid true/false buckets, effectively creating linguistic traps that generate false-positive interventions. By utilizing a multi-class dataset, the LLM successfully isolates these linguistic edge cases, mapping uncertainty to AMBIGUOUS, daily conversations to UNKNOWN, etc. This granularity, despite presenting a mathematically harder classification task, directly prevents inappropriate interventions and drastically improves overall dialogue safety.

Evaluated at a predefined significance level of $\alpha = 0.05$, McNemar's test confirmed statistically significant performance improvements for the LLM system on both datasets ($p < 0.001$). These results

reject $H_{0,4}$, as the LLM handled ambiguous responses, non-standard phrasing, and temporal context shifts significantly better than rigid keyword rules.

A post-hoc statistical power analysis was also conducted for McNemar's test. In this specific test, statistical power is driven by the number of disagreements between the models, meaning the exact cases where one model guessed correctly but the other failed. On the SAD benchmark dataset, the models disagreed on 135 cases, and on the custom conversational dataset, they disagreed on 112 cases. Because the models had a high number of disagreements, the calculated statistical power for both datasets reached 100%. This confirms that the test contained enough examples to fully accept the results, proving that the LLM is superior to the keyword system.

6.3.4 User Acceptance and Natural Language Explainability

RQ5: Can an uncertainty-based query with a natural language explanation of the physiological prediction increase user acceptance compared to unexplained queries?

- **Null Hypothesis ($H_{0,5}$):** Providing a natural language explanation for a physiological prediction (e.g., explaining why a query was triggered) does not significantly alter user acceptance in the system.
- **Alternative Hypothesis ($H_{1,5}$):** Users exposed to inquiries accompanied by natural language explanations of physiological triggers report significantly higher acceptance compared to users exposed to unexplained queries.

Addressing RQ5, the user study (N=11) evaluated participant perception of uncertainty-driven queries containing natural language explanations. Table 6.5 presents the Likert scale responses for explainability metrics (QXAI_01 to QXAI_04).

Table 6.5: Participant evaluation of query explainability (1 = Strongly Disagree, 5 = Strongly Agree).

Statistic	**QXAI_01**	**QXAI_02**	**QXAI_03**	**QXAI_04**
Mean	4.00	4.09	4.18	4.27
Median	4.00	4.00	4.00	4.00
Mode	4	4	4	4
Standard Deviation	0.63	0.54	0.87	0.65
Minimum	3	3	2	3
Maximum	5	5	5	5

Across all criteria, mean evaluation scores exceeded 4.00 out of 5.00. In the direct preference task (QXAI), 72.7% of participants ($n = 8$) preferred queries provided with natural language explanations (Option B) compared to 27.3% ($n = 3$) who preferred unexplained queries (Option A). These results reject $H_{0,5}$, demonstrating that natural language explanations significantly improve user acceptance.

7

Conclusions

7.1 Contributions

- Developed a real-time, closed-loop ROS 2 framework for stress inference capable of handling both WESAD dataset simulations and live EmotiBit hardware streams.
- Implemented an uncertainty-aware Active Learning algorithm that blends a static Random Forest model with a personalized model, dynamically updating weights based on human feedback.
- Integrated a localized Qwen2.5:3B LLM to perform semantic dialogue management, categorizing user intent into eight distinct behavioral tags.
- Conducted a pilot user study based on SHAP feature attributions to evaluate user preference for transparent natural language feedback versus unexplained queries.

7.2 Lessons learned

- Transitioning to live hardware streaming introduces a higher processing latency (295.50 ms) compared to offline pre-processed data (6.14 ms), but it remains entirely viable for real-time operation and falls well within practical equivalence margins.
- Redundant interruptions during prolonged physiological episodes only decrease when the personalized active learning model is assigned a dominant blend weight ($w \geq 0.9$).
- Model classification accuracy and F1-score plateau at an adaptation weight of w = 0.7 due to the finite number of labeled subject samples available within the WESAD dataset.
- An LLM-based intent classifier is significantly more robust at handling ambiguous user responses, physical symptoms, and temporal context shifts than rigid keyword-based systems.
- Users demonstrate a strong preference (72.7%) for system interventions that provide detailed, natural language explanations of the underlying physiological data triggering the alert.

7.3 Future work

- Fully integrate SHAP directly into the real-time system to dynamically generate natural language explanations for physiological predictions.
- Future iterations should evaluate the adaptive framework on larger, more diverse longitudinal datasets to overcome the personalization plateau observed with the WESAD constraints.
- Conduct comprehensive offline user studies to validate ground truth accuracy, for the real-time hardware streams.
- Further research can explore the integration of self-supervised representation learning techniques, such as Contrastive Learning, to better handle raw signal noise and motion artifacts prior to user interaction.

A

Design Details

- The system architecture operates across four continuous stages: a Data Acquisition Layer, Core Inference and Decision Routing, Execution Pathways, and an Interaction/Continuous Learning layer.
- Signal processing utilizes double-ended queues (deque) to manage 60-second sliding windows, preventing memory bloat during continuous physiological streaming.
- Online model retraining is handled asynchronously in a background thread to prevent blocking the primary ROS 2 node during active learning updates.
- The dialogue interaction history is constrained to a rolling buffer of the six most recent conversational exchanges, optimizing the context-window efficiency for the local LLM while preserving necessary interaction memory.
- The Live Hardware ingestion node utilizes multithreading to pull from multiple LSL channels EDA and PPG concurrently, preventing synchronization stalls and dropped sensor frames.
- The personalized Active Learning classifier relies on a 50-estimator Random Forest model configured with balanced class weights to mitigate the inherent class imbalances typical in sparsely labeled human-in-the-loop data.
- The offline dataset publisher incorporates auto-jump logic to algorithmically bypass unlabelled transitions, automatically fast-forwarding to relevant stress and calm episodes to streamline simulated real-time evaluation.
- The LLM intent classifier enforces a strict generation temperature of 0.0 to guarantee deterministic, highly reproducible semantic classification.

B

Parameters

- The system's prediction uncertainty threshold is defined as a Shannon entropy value greater than 0.45 ($H > 0.45$).
- The state machine enforces a 180-second Uncertainty Streak before triggering an active learning query.
- Intervention cooldown periods are configured to 300 seconds for both Calm and Stress states to prevent user alert fatigue.
- Automated breathing exercises are programmed to execute for 4 total cycles, with each cycle lasting 16 seconds.
- The sliding window processor ingests BVP data at 64 Hz and EDA data at 4 Hz, triggering feature extraction across the 60-second observation windows.
- The maximum wait time for user feedback during an active interaction window is set to 120 seconds, after which a gentle follow-up reminder is dispatched.
- If no user response is detected 30 seconds after the follow-up reminder, the reply window is automatically abandoned, and all physiological state streaks are reset to zero.
- The dynamic adaptation blend weight (w) scales proportionally based on the volume of collected user feedback, reaching maximum saturation after 20 personalized labels are acquired.
- The minimum number of manual labels required to trigger the first background retraining of the personalized active learning model is 2.

C

Description of AI-Generated Content

All AI-generated content was critically reviewed, and the author takes full responsibility for the careful scientific work and final text presented.

- Grammerly:

  - To enhance the overall readability, flow, and grammatical accuracy of the final written report.

- Google Gemini :

  - To generate a robust, diverse, and domain-specific custom dataset required for evaluating the LLM classification system.

  - To brainstorm, test, and refine the system instructions and prompts used to guide the classification behavior of the LLM.

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